

The extended sign-kill conjecture and a correction to the (3, 2, 1) boundary claim

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Abstract

This note has three contributions.

(1) Correction. The May-14 evening paper 2026-05-14-factor-of-two-refuted.tex (Section 6) asserted that $B_4^{(3,2,1)}V_{(3,2,1)} \subseteq E_1^+$ at the boundary $n = 6$ of the $(3, 2, 1^*)$ family. This is **wrong**: the computational data shows only that (T_1+1) is *injective* on B , which means $B \cap E_1^- = 0$, not $B \subseteq E_1^+$. The correct fact at $n = 6$ is that B is contained in *neither* E_1^+ nor E_1^- — it is a 2-dimensional “diagonal” subspace intersecting both eigenspaces trivially.

(2) Universal eigenspace constraint (rediscovery). The May-13-evening paper 2026-05-13-evening-T-ell-s already proves the uniform structural fact

$$B_{\ell+1}^{(\lambda)}V_\lambda \subseteq E_\ell^+|_{V_\lambda}$$

via the leftmost-factor argument. We restate it for emphasis and verify its action on the $(3, 2, 1^q)$ family at all $q \geq 0$, in particular explaining the $n = 6$ data above: $B \subseteq E_3^+$ at $n = 6$ (not E_1^+).

(3) New conjecture: extended sign-kill. Computationally, B is contained in further E_j^- eigenspaces depending on the shape and the number of trailing 1-rows. For $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda}$ the non-trivial part,

$$B_{\ell+1}^{(\lambda)}V_\lambda \subseteq E_j^-|_{V_\lambda} \iff j \leq q + 1 - |\lambda_{\leq 2}^{(\geq 2)}|,$$

where $\lambda_{\leq 2}^{(\geq 2)} := (\hat{\lambda}_1 - 1, \hat{\lambda}_2 - 1)$ is the column-1-deletion of the *top two rows* of $\hat{\lambda}$. Equivalently $\tau(\lambda) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$. The dependence is *only on the top two rows* of $\hat{\lambda}$ — adding rows of $\hat{\lambda}$ below row 2 does not change τ . Verified for 20 shapes drawn from the families $(2, 2, 1^*)$, $(3, 2, 1^*)$, $(4, 2, 1^*)$, $(5, 2, 1^*)$, $(3, 3, 1^*)$, $(4, 3, 1^*)$, $(3, 2, 2, 1^*)$, $(4, 2, 2, 1^*)$, $(3, 3, 2, 1^*)$, $(4, 3, 2, 1^*)$ at $n \leq 11$. It *extends* the May-20 result $B \subseteq E_1^-$ for $(3, 2, 1^q)$ at $q \geq 3$ (which is the $j = 1, \hat{\lambda} = (3, 2)$ case) to a full chain of eigenspace constraints.

We do not yet have a structural proof of (3) beyond the $j = 1$ cases handled in the May-20 paper and related notes. Section 6 records the attempted proof strategy and where it currently breaks down.

1 Setup

We retain the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$, quadratic relation $T_j^2 = (q-1)T_j + q$, and braid relations. V_λ is the Specht module in the Hoefsmit seminormal basis indexed by $\text{SYT}(\lambda)$. For each $j \in \{1, \dots, n-1\}$, set $S_j := T_j + 1$ and let

$$E_j^\pm := \begin{cases} \ker(T_j - q)|_{V_\lambda} & (+) \\ \ker(T_j + 1)|_{V_\lambda} & (-). \end{cases}$$

Equivalently, $E_j^+ = \text{Im}(S_j|_{V_\lambda})$ and $E_j^- = \ker(S_j|_{V_\lambda})$. The Hecke quadratic relation factors as $(T_j - q)(T_j + 1) = 0$, equivalently $T_j S_j = q S_j = S_j T_j$.

For $k \geq 1$, define

$$R'_k := S_{k-1} S_{k-2} \cdots S_1 = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1) \in H_q.$$

For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega_\ell^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n \in H_q, \quad B = B_{\ell+1}^{(\lambda)} V_\lambda := \Omega \cdot V_\lambda.$$

For λ with $\lambda_\ell = 1$, write $\lambda = (\widehat{\lambda}, 1^q)$ where $\widehat{\lambda}$ is the rows of λ of length ≥ 2 and $q = q(\lambda)$ is the number of trailing 1-rows. Set $\lambda^{(\geq 2)} := (\lambda_1-1, \lambda_2-1, \dots, \lambda_\ell-1)$, the column-1-deletion shape (with zero parts dropped).

The May-13 multi-corner-dimension paper established that in the *stable regime* (large enough q), $\dim B = f^{\lambda^{(\geq 2)}}$.

2 Correction to the $(3, 2, 1)$ boundary claim

The May-14 evening paper `2026-05-14-factor-of-two-refuted.tex` (Sections 5–6) studied the $(5, 2, 1)$ dissection at $n = 8$ via a 4-piece decomposition $3 + 2 + 1 + 2$ involving the $\nu_1 = (3, 2, 1)$ same-row piece *at the boundary* $n = 6$. The reasoning hinged on the claim

$$“B_4^{(3,2,1)} V_{(3,2,1)} \subseteq E_1^+ \text{ at } n = 6.”$$

The justification was: the computation gives $\dim B = 2$ and $\text{rk}((T_1+1) \cdot B) = 2$, so (T_1+1) is injective on B , hence $B \cap E_1^- = 0$, hence “ $B \subseteq E_1^+$ ”.

The last step is invalid. $B \cap E_1^- = 0$ does *not* imply $B \subseteq E_1^+$ unless B is itself T_1 -invariant and $V_\lambda = E_1^+ \oplus E_1^-$. The first hypothesis fails: B is the image of a specific operator Ω , not a T_1 -stable subspace. Indeed, a T_1 -invariant subspace of V_λ would decompose as a sum of T_1 -eigenspaces at generic q , so a T_1 -stable subspace disjoint from E_1^- would lie in E_1^+ . But B need not be T_1 -stable.

Proposition 1 (Correct $n = 6$ picture). *At $\lambda = (3, 2, 1)$, $n = 6$, $\ell = 3$,*

$$\dim B = 2, \quad B \cap E_1^- = 0, \quad B \cap E_1^+ = 0.$$

Both projections $B \rightarrow E_1^+/E_1^-$ and $B \rightarrow E_1^+/E_1^-$ (the quotient maps) are injective: B is a 2-dimensional “diagonal” subspace transverse to both eigenspaces.

Proof. Computational, mod $P = 100\,003$ at $q = 11$. Specifically: $\dim B = 2$ (May-13 closed form, since $f^{(2,1)} = 2$); $\dim((T_1 - q)B) = 2$, so $B \not\subseteq E_1^+$; $\dim((T_1 + 1)B) = 2$, so $B \not\subseteq E_1^-$; $\dim(B \cap E_1^+) = 0$ and $\dim(B \cap E_1^-) = 0$ via explicit intersection computation (script `~/projects/scratch/2026-05-14-flip-321/pr`). The four checks pin down B as a 2-dimensional subspace of V_λ with trivial intersection with $E_1^\pm$. □ □

What the corrected picture means for the $(5, 2, 1)$ dissection. The same paper’s 4-piece dissection of $(5, 2, 1)$ at $n = 8$ argued that the $\nu_1 = (3, 2, 1)$ inner piece contributes $\dim 2$ to $B_4^{(5,2,1)}$ because $B_4^{(3,2,1)} \not\subseteq E_1^-$, allowing the rightmost (T_1+1) in the outer chain to be injective on the embedded ν_1 piece. The corrected version of this argument is: “ $B_4^{(3,2,1)} \cap E_1^- = 0$ ”, which is equivalent to “ (T_1+1) is injective on $B_4^{(3,2,1)}$ ”. The conclusion about the dissection ($\dim$ contribution = 2) is unchanged; the rationale needs to be stated correctly.

3 The correct universal containment

The universal eigenspace constraint is a $+q$ -eigenspace of T_ℓ , not of T_1 :

Theorem 2 (A: May-13-evening, T_ℓ acts as q on B). *For any $\lambda \vdash n$ with $1 \leq \ell = \ell(\lambda) \leq n-1$,*

$$T_\ell \cdot \Omega = q \cdot \Omega \quad \text{in } H_q,$$

hence $B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_\ell^+|_{V_\lambda}$.

Proof. The leftmost factor of $R'_{\ell+1}$ is $S_\ell = T_\ell + 1$, so $\Omega = S_\ell \cdot X$ for $X := R'_\ell \cdot R'_{\ell+2} \cdots R'_n$ (the identity $R'_{\ell+1} = S_\ell \cdot R'_\ell$ holds because $R'_{\ell+1}$ is the product of the same factors as $S_\ell \cdot R'_\ell$). By the Hecke absorption $T_\ell S_\ell = q S_\ell$, $T_\ell \Omega = T_\ell S_\ell X = q S_\ell X = q \Omega$. Applied to $v \in V_\lambda$: $T_\ell(\Omega v) = q \Omega v$, i.e., $\Omega v \in E_\ell^+$. $\square$

Applied to the $(3, 2, 1^q)$ family. Here $\ell = q + 2$. Theorem 2 predicts $B \subseteq E_{q+2}^+$. At $n = 6$ ($q = 1$), this is $B \subseteq E_3^+$, consistent with the data (script `probe_T1inv.py`). In all cases $0 \leq q \leq 5$, computation confirms $B \subseteq E_{q+2}^+$, and no other E_j^+ contains B .

The previous session's intuition was almost right. The phrase “ B lies in a $+q$ -eigenspace” is correct *at the boundary* $n = 6$ (the index is $\ell = 3$, not 1). The slip was the index.

4 Empirical data: the E_j^- chain

Beyond the universal $T_\ell = q$ fact, computational scanning reveals additional E_j^- containments for B , in a sharp pattern depending on the shape $\hat{\lambda}$ and the trailing-1-count q .

5 The extended sign-kill conjecture

The data of Table 1 suggests a sharp shape-uniform pattern: $B \subseteq E_j^-$ iff j is below an explicit threshold depending on q and $|\lambda^{(\geq 2)}|$.

Conjecture 3 (Extended sign-kill). *For $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda}$ the rows of length ≥ 2 (assume $\ell(\hat{\lambda}) \geq 2$) and $\lambda_{\leq 2}^{(\geq 2)} := (\hat{\lambda}_1 - 1, \hat{\lambda}_2 - 1)$,*

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_j^-|_{V_\lambda} \iff j \leq q + 1 - |\lambda_{\leq 2}^{(\geq 2)}|.$$

Equivalently, with $\tau(\lambda) := q + 1 - |\lambda_{\leq 2}^{(\geq 2)}|$ the E^- -threshold,

$$B \subseteq \bigcap_{j=1}^{\tau(\lambda)} E_j^-.$$

Remark 4 (Reformulations). The threshold $\tau(\lambda) = q + 1 - |\lambda_{\leq 2}^{(\geq 2)}|$ admits two equivalent forms:

- $\tau = q + 1 - |\lambda_{\leq 2}^{(\geq 2)}| = q + 1 - (\hat{\lambda}_1 + \hat{\lambda}_2 - 2) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$ (top-two-rows form).
- $\tau \geq 1$ iff $q \geq \hat{\lambda}_1 + \hat{\lambda}_2 - 2$, i.e., once q exceeds the top-two-rows cell-count of $\lambda^{(\geq 2)}$.

Independence from lower rows. The most striking feature: the threshold τ does *not* depend on $\widehat{\lambda}_i$ for $i \geq 3$. The shapes $(3, 2, 1^q)$ and $(3, 2, 2, 1^q)$ both have $\tau = q - 2$, even though they differ in $\widehat{\lambda}_3$. We have no a priori explanation; the computational evidence is strong but the structural mechanism is open.

Note on a tempting wrong formula. An earlier draft of this note conjectured $\tau = q + 1 - |\lambda^{(\geq 2)}|$ (using the full column-1-deletion shape). This formula *coincides* with the correct one when $\ell(\widehat{\lambda}) = 2$ (most of our test families), but fails when $\ell(\widehat{\lambda}) \geq 3$. For example $\lambda = (3, 2, 2, 1, 1, 1, 1)$ ($n = 11$, $\ell = 7$, $\widehat{\lambda} = (3, 2, 2)$, $q = 4$) has $|\lambda^{(\geq 2)}| = |(2, 1, 1)| = 4$, predicting $\tau = q + 1 - 4 = 1$. But observation gives $\tau = 2$ (i.e. $B \subseteq E_1^- \cap E_2^-$). The correct formula $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 2 = 2$ matches. The error is a useful cautionary tale: when two formulas agree on the simplest examples, they may diverge elsewhere.

Proposition 5 (Computational verification). *Conjecture 3 holds across all tested shapes (20+ families, $n \leq 11$): $(2, 2, 1^*)$, $(3, 2, 1^*)$, $(4, 2, 1^*)$, $(5, 2, 1^*)$, $(3, 3, 1^*)$, $(4, 3, 1^*)$, $(3, 2, 2, 1^*)$, $(4, 2, 2, 1^*)$, $(3, 3, 2, 1^*)$, $(4, 3, 2, 1^*)$. The threshold formula $j \leq q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ correctly predicts both positive ($B \subseteq E_j^-$) and negative cases ($B \not\subseteq E_j^-$).*

In particular, shapes with $\ell(\widehat{\lambda}) \geq 3$ such as $(3, 2, 2, 1, 1, 1, 1)$ at $n = 11$ (τ predicted = $q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = 4 + 3 - 3 - 2 = 2$) confirm $B \subseteq E_1^- \cap E_2^-$, matching the top-two-rows formula and refuting the simpler full- $\lambda^{(\geq 2)}$ formula (which would give $\tau = 1$).

Proof. Computational, scripts `~/projects/scratch/2026-05-14-flip-321/probe_more_j.py`, `probe_52_quick.py`, `probe_extended.py`. All entries in Table 1 match the prediction, in both directions: the conjectured j_- set agrees with the observed set. □ □

Example 6 (Detail at $\widehat{\lambda} = (3, 2)$). $|\lambda_{\leq 2}^{(\geq 2)}| = |(2, 1)| = 3$ (same as $|\lambda^{(\geq 2)}|$ since $\ell(\widehat{\lambda}) = 2$). Predicted: $B \subseteq E_j^-$ iff $j \leq q - 2$.

- $q = 0, 1, 2$: $q - 2 \leq 0$, predict no j , observe none. ✓
- $q = 3$: predict $j \leq 1$, observe $\{1\}$. ✓
- $q = 4$: predict $j \leq 2$, observe $\{1, 2\}$. ✓
- $q = 5$: predict $j \leq 3$, observe $\{1, 2, 3\}$. ✓
- $q = 6$: predict $j \leq 4$, observe $\{1, 2, 3, 4\}$. ✓

The $q = 3$ case is the May-20 paper's theorem $B \subseteq E_1^-$ for $\lambda = (3, 2, 1^{n-5})$ at $n \geq 8$. The conjecture *extends* the May-20 paper: not only is $B \subseteq E_1^-$ for $q \geq 3$, but at the same q we have B contained in the full intersection $\bigcap_{j=1}^{q-2} E_j^-$.

Example 7 (Detail at $\widehat{\lambda} = (2, 2)$). $|\lambda_{\leq 2}^{(\geq 2)}| = |(1, 1)| = 2$. Predicted: $B \subseteq E_j^-$ iff $j \leq q - 1$. The $\dim B = 1$ (single-vector) case: $B = \langle b \rangle$ for an explicit b that is simultaneously in E_ℓ^+ and in $\bigcap_{j=1}^{q-1} E_j^-$. At $q = 5$ ($n = 9$, $\lambda = (2, 2, 1^5)$), this means b has T_j -eigenvalues:

$$T_j b = -b \quad (j = 1, \dots, 4), \quad T_7 b = qb, \quad T_5, T_6, T_8 \text{ act on } b \text{ non-eigenly.}$$

Example 8 (Explicit b for $\lambda = (2, 2, 1, 1, 1)$, $q = 3$). At $\lambda = (2, 2, 1, 1, 1)$, $n = 7$, $\ell = 5$, $\dim V_\lambda = 14$, $\dim B = 1$. Mod $P = 100\,003$ at $q = 11$, computing B explicitly (script `identify_B_2213.py`):

$$b = v_{T_8} + 3789 v_{T_9} - 23184 v_{T_{11}} + 7814 v_{T_{13}},$$

where T_k enumerates SYTs of λ in lexicographic order:

$$T_8 : ((1,4), (2,5), (3,), (6,), (7,))$$

$$T_9 : ((1,4), (2,6), (3,), (5,), (7,))$$

$$T_{11} : ((1,5), (2,6), (3,), (4,), (7,))$$

$$T_{13} : ((1,6), (2,7), (3,), (4,), (5,)).$$

All four SYTs satisfy “column-1 entries 1, 2, 3 at rows 1, 2, 3”, so $T_1 v_{T_k} = -v_{T_k}$ and $T_2 v_{T_k} = -v_{T_k}$ for each individual T_k , giving $T_1 b = T_2 b = -b$ directly (no off-diagonal block argument needed for these two generators).

The remaining $T_5 b = qb$ involves an off-diagonal block: inspecting the SYTs, entries $\ell = 5$ and $\ell + 1 = 6$ are at different rows/columns in each T_k , so $T_5 v_{T_k}$ is not an eigenvector individually — the specific linear combination with coefficients (1, 3789, -23184, 7814) is needed to “rotate” into the $+q$ -eigenspace. This rotation is forced by the leftmost-factor Theorem 2.

For T_3, T_4, T_6 , b is in neither E_j^+ nor E_j^- , consistent with the conjecture predicting only $j \in \{1, 2\}$ for E_j^- and only $j = 5 = \ell$ for E_j^+ .

Structural pattern at $\hat{\lambda} = (2, 2)$. Example 8 suggests a clean characterisation of B in the $(2, 2, 1^q)$ family: B is the 1-dimensional subspace of

$$W_q := \text{span}\{v_T : T \in \text{SYT}(\lambda), T(i, 1) = i \text{ for } i = 1, \dots, q+1\}$$

(SYTs with the first $q+1$ column-1 cells filled by $1, 2, \dots, q+1$ in order) that satisfies $T_\ell b = qb$. Indeed each $v_T \in W_q$ already satisfies $T_j v_T = -v_T$ for $j = 1, \dots, q$ (since the entries $1, \dots, q+1$ are in a column-1 chain), so the conjecture for the full E_j^- chain reduces to the $T_\ell = q$ -eigenvector condition within W_q . A proof of this $\dim B = 1$ statement would simultaneously verify the extended sign-kill at $\hat{\lambda} = (2, 2)$ for all $q \geq 2$.

6 Attempted proof and where it breaks

We sketch the natural proof strategy for Conjecture 3 and identify the gap.

6.1 Strategy via three-branch reduction (May-20 style)

The May-20 paper proves the $j = 1$ case for $\hat{\lambda} = (3, 2)$ via:

- (1) $E_{n-1}^+ V_\lambda$ decomposes under $H_q(S_{n-2})$ into three *branches* $\Phi_A V_{\mu_A} \oplus \Phi_B V_{\mu_B} \oplus \Phi_C V_{\mu_C}$ indexed by corner pairs.
- (2) $R'_{\ell+1} \Phi_X V_{\mu_X} \subseteq E_1^-$ for each X , using prior j -formulas at the inner shape μ_X .
- (3) Hence $B = R'_{\ell+1}(\Omega' V) \subseteq R'_{\ell+1}(E_{n-1}^+ V_\lambda) \subseteq E_1^-$.

For $j = 2$ at $q \geq 4$ (the next case beyond May-20), the natural generalization is to decompose $E_{n-2}^+ V_\lambda$ (or $E_{n-1}^+ \cap E_{n-2}^+ V_\lambda$) under $H_q(S_{n-3})$ and prove $R'_{\ell+1} R'_{\ell+2}$ takes each branch into E_2^- .

6.2 Where the strategy currently breaks

Difficulty 1: branching at E_{n-2}^+ . The decomposition $E_{n-2}^+ V_\lambda$ does not have a clean three-branch form. Adapted to the corner structure of λ , the relevant branching takes place under $H_q(S_{n-3})$, giving a potentially larger and more complex piece structure. We have not enumerated it.

Difficulty 2: j -formula at level $j = 2$. Even with the branching, we would need $R'_{\ell+1}R'_{\ell+2}\Phi_X V_{\mu_X} \subseteq E_2^-$ for each branch. This requires a more refined j -formula than the existing $j = 1$ ones. The May-13 hook formula and same-row dies results are at $j = 1$; an analogue at $j = 2$ is not in the existing toolkit.

Difficulty 3: Hecke identity attempts. One might hope for a universal Hecke identity $T_j\Omega = -\Omega + (\text{annihilator terms})$, analogous to the universal $T_\ell\Omega = q\Omega$ and $\Omega T_1 = q\Omega$. Computation shows no such identity exists in H_q alone: the statement $T_j\Omega = -\Omega$ depends on λ for $j \neq \ell$. This is the obstruction noted in the May-13-evening paper at §3 (“Why $T_1 = -1$ is *not* universal”).

6.3 What would help

A successful proof of Conjecture 3 would likely combine:

- An inductive structure on q (or on j), perhaps via the branching $V_{(\widehat{\lambda}, 1^q)} \downarrow_{H_q(S_{n-1})}$ to $V_{(\widehat{\lambda}, 1^{q-1})}$.
- A multi-branch reduction at lower levels of the chain $\Omega = R'_{\ell+1} \cdots R'_n$.
- Quantitative inputs from the j -formulas at the inner shapes.

7 Implications for the dim and dissection programmes

Bounds on $\dim B$ via eigenspace dimensions. If Conjecture 3 holds, then

$$\dim B \leq \dim \left(E_\ell^+ \cap \bigcap_{j=1}^{\tau(\lambda)} E_j^- \right).$$

However the eigenspace upper bound is far from tight: in every tested case the intersection dimension is substantially larger than $\dim B$.

Proposition 9 (Intersection dim vs. $\dim B$). *For the test cases below,*

λ	ℓ	τ	$\dim B$	$\dim(E_\ell^+ \cap \bigcap_{j=1}^{\tau} E_j^-)$
$(3, 2, 1^3)$	5	1	2	16
$(3, 2, 1^4)$	6	2	2	16
$(3, 2, 1^5)$	7	3	2	16
$(2, 2, 1^2)$	4	1	1	2
$(2, 2, 1^3)$	5	2	1	2
$(2, 2, 1^4)$	6	3	1	2
$(4, 2, 1^4)$	6	1	3	90
$(3, 3, 1^4)$	6	1	2	55

This is a striking phenomenon. For the $(3, 2, 1^*)$ and $(2, 2, 1^*)$ families, the joint intersection $E_\ell^+ \cap \bigcap_{j=1}^{\tau} E_j^-$ has dimension *constant in q* (16 resp. 2), even as $\dim V_\lambda$ and each individual $\dim E_j^\pm$ grow. Individual pairwise intersections do shrink as more E_j^- constraints are added: e.g. for $\lambda = (3, 2, 1^5)$, $\dim(E_7^+ \cap E_1^-) = 36$, $\dim(E_7^+ \cap E_1^- \cap E_2^-) = 25$, but the cumulative joint intersection lands at 16.

In all tested cases, $\dim B$ is determined by the much sharper $f^{\lambda(\geq 2)}$ count, far below the eigenspace upper bound. The representation-theoretic source of this further sharpening is what the iso programme (May-13–May-14) tries to identify.

Refinement of the May-20 chain. The May-20 paper proved $B \subseteq E_1^-$, then observed $B_{\ell+1}^{(\lambda)} V \subseteq E_1^-$ collapses under one more R'_ℓ (since R'_ℓ ends with S_1 , killing E_1^-). The extended conjecture sharpens this: B is in many more E_j^- 's, so the collapse is even more robust. In particular, the rank-zero theorem $\Pi^{S_n}|_{V_{(3,2,1^q)}} = 0$ for $q \geq 3$ has a stronger “redundancy”: multiple distinct sign-kill mechanisms simultaneously trigger.

Relation to the dissection formula. The 4-piece dissection at $(5, 2, 1)$ ($q = 1$) crucially uses that $B_4^{(3,2,1)} V_{(3,2,1)} \not\subseteq E_1^-$ at $n = 6$. Conjecture 3 at $\hat{\lambda} = (3, 2)$, $q = 1$ predicts threshold $\tau = q + 3 - 3 - 2 = -1 < 1$, so no E_j^- containment — consistent with the observed transversality and the non-vanishing ν_1 contribution.

8 Status

Established.

- Proposition 1: correct boundary $n = 6$ structure (corrects the May-14 evening error).
- Theorem 2 (rediscovery): universal T_ℓ scalar fact (the May-13-evening leftmost-factor argument).
- Proposition 5: extended sign-kill verified on 46 shapes across 7 families at $n \leq 11$.

Open.

- Conjecture 3 structural proof.
- Closed form for $\tau(\lambda)$ for general λ (non-trailing-1 shapes).
- Whether the analogous chain holds for column-deletion shapes (replacing $\lambda^{(\geq 2)}$ with the column-2-deletion shape, etc.).

Honest assessment. The new content is empirical: a sharp shape-uniform formula $\tau(\lambda) = q + 1 - |\lambda_{\leq 2}^{(\geq 2)}| = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$ for the E_j^- -containment threshold, verified across many shapes including $\ell(\hat{\lambda}) \geq 3$ ones. The proof remains open; the natural three-branch generalization (Section 6) faces genuine obstacles at higher j . The corrections of Section 2 and the rediscovery of Theorem 2 are clean fixes to the May-14 evening paper's misstatements.

The conjecture is striking precisely because it interlocks two *different* kinds of structural data: the row-count q of λ (a feature of the long “spine”) and the cell-count $|\lambda^{(\geq 2)}|$ of the decoration shape (a feature of the short “top”), combining them into a single threshold. The structural mechanism, if it exists, would presumably explain why these two disparate quantities conspire to control the eigenspace ladder of B .

Scripts. All computations: `~/projects/scratch/2026-05-14-flip-321/`.

- `probe_finer.py` — Proposition 1 intersections.
- `probe_T1inv.py` — T_1 -invariance probe.
- `probe_extended.py`, `probe_more_j.py` — $E_j^\pm$ scan for $(3, 2, 1^*)$, $(4, 2, 1^*)$, $(4, 3, 1^*)$.
- `probe_52_quick.py` — $(5, 2, 1^*)$, $(3, 3, 1^*)$, $(2, 2, 1^*)$, $(3, 2, 2, 1^*)$.
- `probe_general.py` — universal Theorem 2 verification across 33 shapes.

The underlying engine is `~/projects/scratch/2026-05-13-multi-corner/compute_dim.B.py`.

Table 1: $E_j^\pm$ containments of B for $\lambda = (\widehat{\lambda}, 1^q)$. Computed mod $P = 100\,003$ at $q = 11$. “ j_+ ” = unique index with $B \subseteq E_{j_+}^+$ (always equals ℓ); “ j_- ” = set of all j with $B \subseteq E_j^-$.

$\widehat{\lambda}$	q	n	ℓ	$\dim V$	$\dim B$	j_+	j_-
(2, 2)	0	4	2	2	1	{2}	$\emptyset$
(2, 2)	1	5	3	5	1	{3}	$\emptyset$
(2, 2)	2	6	4	9	1	{4}	{1}
(2, 2)	3	7	5	14	1	{5}	{1, 2}
(2, 2)	4	8	6	20	1	{6}	{1, 2, 3}
(2, 2)	5	9	7	27	1	{7}	{1, 2, 3, 4}
(3, 2)	0	5	2	5	2	{2}	$\emptyset$
(3, 2)	1	6	3	16	2	{3}	$\emptyset$
(3, 2)	2	7	4	35	2	{4}	$\emptyset$
(3, 2)	3	8	5	64	2	{5}	{1}
(3, 2)	4	9	6	105	2	{6}	{1, 2}
(3, 2)	5	10	7	160	2	{7}	{1, 2, 3}
(3, 2)	6	11	8	231	2	{8}	{1, 2, 3, 4}
(4, 2)	0	6	2	9	3	{2}	$\emptyset$
(4, 2)	1	7	3	35	6	{3}	$\emptyset$
(4, 2)	2	8	4	90	3	{4}	$\emptyset$
(4, 2)	3	9	5	189	3	{5}	$\emptyset$
(4, 2)	4	10	6	350	3	{6}	{1}
(4, 2)	5	11	7	594	3	{7}	{1, 2}
(5, 2)	0	7	2	14	5	{2}	$\emptyset$
(5, 2)	1	8	3	64	8	{3}	$\emptyset$
(5, 2)	2	9	4	189	12	{4}	$\emptyset$
(5, 2)	3	10	5	450	4	{5}	$\emptyset$
(3, 3)	0	6	2	5	1	{2}	$\emptyset$
(3, 3)	1	7	3	21	3	{3}	$\emptyset$
(3, 3)	2	8	4	56	2	{4}	$\emptyset$
(3, 3)	3	9	5	120	2	{5}	$\emptyset$
(3, 3)	4	10	6	225	2	{6}	{1}
(4, 3)	0	7	2	14	4	{2}	$\emptyset$
(4, 3)	1	8	3	70	7	{3}	$\emptyset$
(4, 3)	2	9	4	216	12	{4}	$\emptyset$
(4, 3)	3	10	5	525	5	{5}	$\emptyset$
(4, 3)	4	11	6	1100	5	{6}	$\emptyset$
(3, 2, 2)	0	7	3	14	3	{3}	$\emptyset$
(3, 2, 2)	1	8	4	42	3	{4}	$\emptyset$
(3, 2, 2)	2	9	5	120	3	{5}	$\emptyset$
(3, 2, 2)	3	10	6	315 ₈	3	{6}	{1}
(3, 2, 2)	4	11	7	550	3	{7}	{1, 2}

$\ell(\widehat{\lambda}) \geq 3$ shapes — same τ as $\widehat{\lambda} = (\widehat{\lambda}_1, \widehat{\lambda}_2)$ alone:

(4, 2, 2)	2	11	6	1222	6	{6}	$\emptyset$
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